

BLG Core Validation — Data & Methods Workbook

β -Lactoglobulin Adsorption at the Air–Water Interface
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How to read this document

Every result quoted anywhere in Paper 1 (or in project notes) traces back to one of the entries below. Each entry follows the same structure: what the quantity means and why it was measured, the exact script and command used to produce it, where the input trajectory data and output cache file live, the numeric result actually read out of that cache file (not retyped from memory), and a short interpretation. Re-running any `srccmd` block exactly as written reproduces the number in the results table. All commands assume:

- Working directory: `~/Workspace/MILK_FROTHING/`
- Environment active: `conda activate research-env` (Python 3.12, MDAnalysis 2.10.0), or GROMACS sourced via `source ~/opt/gromacs-2020.4/bin/GMXRC` for scripts that shell out to `gmx`.
- `--label` selects which of the four trajectories to run: `CENTER`, `R1`, `R2`, `R3`, or `all`.

1 Simulation setup

What: Four independent 1000 ns all-atom MD trajectories of BLG (PDB 1BEB) at a slab air–water interface: one from the paper’s original starting structure (`CENTER`) and three replicas with independent velocity seeds (`R1/R2/R3`), 4.00 μ s total.

Force field: CHARMM36, TIP3P water, slab geometry with two vacuum interfaces (protein in the water layer, vacuum above and below along *Z*).

Engine: GROMACS 2020.4

```
gmx mdrun -v -deffnm run -ntmpi 1 -ntomp 6
```

Trajectory files: `outputs_BLG/CENTER/MD1000/traj_comp.xtc` (`CENTER`); `outputs_BLG/REPLICA/MD/MD{1,2,3}.xtc` (`R1/R2/R3`); + AMD extension parts (`R1/R2/R3`)

Topology: `outputs_BLG/CENTER/MD1000/md_1000ns.tpr` (`CENTER`); `md.replicaN.tpr` for `R1/R2/R3` — note: the `_ext.tpr` variants are built with a newer GROMACS (tpx version 138) and are unreadable by this machine’s 2020.4 install (tpx 119); all analysis scripts use the base `md.replicaN.tpr` instead (same topology/atom count, safe for post-hoc analysis).

2 Calyx SASA — is the hydrophobic pocket accessible?

What & why: Solvent-accessible surface area of the 9 calyx-lining residues only (the hydrophobic pocket BLG uses to bind ligands). This is the paper’s headline “accessible calyx” metric — does the pocket stay open at the interface, or collapse?

Script: `scripts/analysis/blg_calyx_sasa.py`

```
python -u scripts/analysis/blg_calyx_sasa.py --label all
```

Method: Whole-protein SASA computed per frame with `freesasa`, after `mda.unwrap` (PBC correction — without this, SASA is inflated to 45–62 nm², a project gotcha, see §12), summed over the calyx atom indices only (CALYX_RESIDS = 39, 41, 56, 58, 92, 103, 105, 107, 125; 133 atoms).

Output: `results/analysis/blg_calyx_sasa_{label}.npz` — keys: `time_ns`, `calyx_sasa_nm2`

Replica	N frames	Mean (nm ²)	Std	Min	Max
CENTER	2001	3.807	0.455	2.548	5.577
R1	2003	3.154	0.338	2.087	4.439
R2	2001	3.588	0.319	2.631	4.709
R3	2001	3.360	0.482	1.706	5.079

Interpretation: The calyx never collapses in any of the 4 independent 1000 ns runs — SASA stays in the 1.7–5.6 nm² range throughout, well above zero. This is the evidence behind the paper’s “accessible calyx maintained across every independent replica” claim. This 3.81 nm² CENTER value supersedes an earlier, untraceable 4.22 ± 1.02 nm² figure once quoted in a superseded narrative report — use only the value in this table.

3 RMSD — structural stability by region

What & why: C α RMSD relative to each replica’s own frame 0, computed for four regions: full backbone, the β -sheet barrel (DSSP-derived, not hand-picked), the C-terminal α -helix (residues 130–140), and the 9-residue calyx patch. Answers: does the protein structurally destabilize during interface contact?

Script: `scripts/analysis/blg_rmsd.py`

```
python -u scripts/analysis/blg_rmsd.py --label all
```

Method: Single global C α -backbone least-squares fit per frame (self-referenced to frame 0), sheet/helix/patch RMSD computed on the same rotation via `groupselections` so all four numbers are on a common frame.

Output: `results/analysis/blg_rmsd_{label}.npz` — keys: `time_ns`, `rmsd_backbone`, `rmsd_sheet`, `rmsd_helix`, `rmsd_patch`

Replica	Backbone (nm)	Sheet (nm)	Helix (nm)	Patch (nm)
CENTER	0.229 $\pm$ 0.072	0.149 $\pm$ 0.025	0.121 $\pm$ 0.050	0.198 $\pm$ 0.081
R1	0.238 $\pm$ 0.039	0.148 $\pm$ 0.024	0.144 $\pm$ 0.034	0.347 $\pm$ 0.088
R2	0.206 $\pm$ 0.026	0.120 $\pm$ 0.014	0.110 $\pm$ 0.031	0.254 $\pm$ 0.059
R3	0.268 $\pm$ 0.046	0.140 $\pm$ 0.015	0.155 $\pm$ 0.038	0.276 $\pm$ 0.064

Interpretation: R1’s patch RMSD (0.347 nm) is a real outlier vs. CENTER/R2/R3 (0.20–0.28 nm) — traced (session 2026-08-25) to R1 having 4 of the project’s 6 total long (≥ 10 ns) contact events; the patch RMSD steps up permanently right after R1’s longest event (362–419.5 ns) and never returns to baseline, while calyx SASA (§2) stays in-range throughout. **Conclusion: real conformational rearrangement correlated with sustained interface contact, not noise — do not average this into a flat cross-replica mean in any figure.** Backbone/sheet/helix RMSD are otherwise consistent across all 4 runs — the fold itself is stable; only the calyx patch responds to contact.

4 RMSF — per-residue flexibility

What & why: Root-mean-square fluctuation per residue over each trajectory — which parts of the chain are locally flexible vs. rigid, independent of RMSD’s global-fit view.

Script: `scripts/analysis/blg_rmsf.py` (precompute, one universe at a time to avoid OOM on this 8 GB machine)

```
python -u scripts/analysis/blg_rmsf.py
```

Output: `results/gate_analysis/rmsf_{center,r1,r2,r3}.resids,rmsf.npy` — `resids` and per-residue RMSF (nm) as parallel arrays, 156 residues each.

Replica	Mean RMSF (nm)	Std	Max (nm)	At residue
CENTER	0.141	0.114	0.584	1 (N-terminus)
R1	0.104	0.079	0.542	1 (N-terminus)
R2	0.094	0.068	0.553	1 (N-terminus)
R3	0.124	0.103	0.553	1 (N-terminus)

Interpretation: The N-terminal residue is the single most flexible point in every replica, by a wide margin over the mean — consistent with a free chain terminus, not a red flag. Body-of-protein flexibility (excluding the N-term outlier) is low across all four runs, another line of evidence for a globally stable fold.

5 Radius of gyration (Rg) — overall compactness

What & why: Whether the protein stays globally compact (folded) or expands/unfolds over the trajectory. Locked validation target for CENTER: 1.496 ± 0.009 nm.

Script: `scripts/analysis/blg_rg.py` (wraps `gmx gyrate`)

```
python -u scripts/analysis/blg_rg.py --label all
```

Output: `results/analysis/blg_rg_{label}.npz` — keys: `time_ns`, `rg_nm`, `rg_mean`, `rg_std`

Replica	Rg mean (nm)	Rg std (nm)
CENTER	1.504	0.0224
R1	1.497	0.0093
R2	1.493	0.0079
R3	1.499	0.0122

Interpretation: All four replicas cluster tightly around ≈ 1.50 nm with sub-0.03 nm standard deviation — globally compact and stable in every independent run, no unfolding.

6 Secondary structure (DSSP)

What & why: Per-residue secondary structure (helix/sheet/coil) over time, via MDAnalysis’s built-in STRIDE-based DSSP. Checks whether the fold’s secondary structure content changes during interface contact.

Script: `scripts/analysis/blg_dssp.py`

```
python -u scripts/analysis/blg_dssp.py --label all
```

Output: `results/analysis/blg_dssp-{label}.npz` — keys: `time_ns`, `residue_ids`, `dssp_codes` (per-frame, per-residue), `frac_helix`, `frac_sheet`, `frac_coil`

Replica	Helix %	Sheet %	Coil %
CENTER	11.8	36.7	51.5
R1	11.9	36.0	52.1
R2	12.4	36.8	50.8
R3	10.5	36.6	52.9

Interpretation: Secondary-structure fractions are essentially identical across all 4 replicas (helix 10.5–12.4%, sheet 36.0–36.8%, coil 50.8–52.9%) — the β -barrel fold is preserved throughout; contact with the interface does not denature the protein.

7 Hydrogen bonds

What & why: Counts of protein–protein, protein–water, and interface water–water hydrogen bonds — quantifies how the protein’s H-bond network compares to bulk/interfacial water (used in the Fig 4 BLG-vs-CAS comparison table).

Script: `scripts/analysis/blg_hbonds.py`

```
python -u scripts/analysis/blg_hbonds.py --label CENTER
```

Method: Rewritten 2026-08-25. Protein–protein and protein–water stages now use native `gmx hbond` with two distinct groups (Protein, SOL) instead of MDAnalysis’s `HydrogenBondAnalysis`. This fixed two real bugs in the old code: (1) the old donor auto-guesser missed all 156 backbone amide-N donors (CHARMM36 names the donor H HN, which the guesser didn’t recognize) — silently excluding the entire secondary-structure H-bond network from a number labeled “protein–protein”; (2) the old protein–water term used a combined (`protein`) or (`water`) selection for both donor and acceptor, then subtracted only the protein–protein term, leaving tens of thousands of ordinary bulk water–water H-bonds still counted as “protein–water”. `gmx hbond` also runs $\sim 1000\times$ faster on this system (full 1000 ns in <90 s vs. prior multi-hour, unfinishable runs). Interface water–water is deliberately kept on MDAnalysis (dynamic reselection is scientifically necessary — interface water population changes membership every frame).

Output: `results/analysis/blg_hbonds-CENTER.npz` — keys: `time_ns`, `n_prot_prot`, `n_prot_water`, `n_water_interface`

H-bond group	Mean count/frame	Std
Protein-protein	107.3	5.6
Protein-water	393.2	13.2
Interface water-water	9539.6	328.4

Interpretation: CENTER only — R1/R2/R3 not yet run (script rewrite/validation happened session 2026-08-25). Values above are validated against a standalone smoke test and match exactly; the old pre-rewrite CENTER value (589.1 protein-protein) is quarantined and was never cited anywhere.

8 Density profile

What & why: Mass density of protein and water along the slab’s Z-axis — confirms the slab geometry and interface position are physically sane (protein sits in the water layer, water density approaches bulk $\approx 1000 \text{ kg/m}^3$ away from the interface, vacuum at the box edges).

Script: `scripts/analysis/blg_density.py` (wraps `gmx density`)

```
python -u scripts/analysis/blg_density.py --label all
```

Output: `results/analysis/blg_density- $\{label\}$ .npz` — keys: `z_nm`, `protein_density`, `water_density` (kg/m^3)

Result: Protein density peak $\approx 68.5\text{--}75.3 \text{ kg/m}^3$ across replicas (CENTER 72.5); water bulk density $\approx 1005 \text{ kg/m}^3$ in all 4 — matches expected TIP3P bulk density, confirming the slab and box setup are physically correct.

9 Surface tension

What & why: Instantaneous surface tension from the pressure tensor, cross-checked against the literature TIP3P value as a sanity check on the whole simulation setup (force field, box, thermostat/barostat behavior).

Script: `scripts/analysis/blg_surface_tension.py`

```
python -u scripts/analysis/blg_surface_tension.py --label all
```

Formula: $\gamma = (L_z/2) \times [P_{zz} - (P_{xx} + P_{yy})/2]$, converted from $\text{bar}\cdot\text{nm}$ to mN/m . Cross-checked against GROMACS’s own `#Surf*SurfTen` .edr term for CENTER: matches exactly ($2\times$ the per-interface value, confirming sign/factor convention).

Output: `results/analysis/blg_surface_tension- $\{label\}$ .npz` — keys: `time_ns`, `gamma_mNm`, `gamma_mean`, `gamma_std`

Replica	γ mean (mN/m)	Std
CENTER	51.86	38.52
R1	52.67	38.65
R2	51.98	39.04
R3	52.64	39.40

Interpretation: $\approx 52 \text{ mN/m}$ matches the real TIP3P literature value ($\approx 50\text{--}52 \text{ mN/m}$, Vega & de Miguel 2007) almost exactly across all 4 replicas — a solid cross-check that the force field and box

setup are behaving correctly. Note: an older draft of `main.tex` (§line 144) claims TIP3P is “half” of the experimental 72 mN/m (≈ 35.8 mN/m) — this contradicts the verified 51.9 mN/m CENTER value and is flagged as an error still to be fixed when paper work resumes.

10 Contact detection, gate analysis & SASA–orientation correlation

What & why: The paper’s central mechanistic question: does BLG’s calyx open (“gate”) when the protein contacts the interface, correlating pocket exposure with orientation? This was the original two-factor gate hypothesis.

Scripts: `scripts/analysis/blg_contact.py` (nearest-atom contact detector, flags contact when any protein heavy atom comes within 0.3 nm of the interface); `scripts/analysis/blg_gate_analysis.py` (per-frame SASA + calyx orientation angle + nearest-atom distance, PBC-corrected via `mda.unwrap`); `blg_bootstrap.py` (block-bootstrap CI for Pearson r , accounting for trajectory autocorrelation); `blg_verify_events.py` (per-event SASA/angle stats for the 6 long (≥ 10 ns) contact events).

```
python detect_adsorption_contact.py --tpr <file.tpr> --xtc <file.xtc> \
  --label CENTER --out results/figures/
python scripts/gate_analysis_all_replicas.py
python scripts/analysis/blg_bootstrap.py
python scripts/analysis/blg_verify_events.py
```

Output: `results/gate_analysis/{CENTER,R1,R2,R3}_gate.npz` — keys: `time`, `sasa`, `angle`, `min.dist`

Replica	SASA mean (nm ²)	SASA std	Angle mean (°)	Min dist mean (nm)
CENTER	28.93	2.77	93.7	0.702
R1	29.08	1.21	102.1	0.577
R2	27.86	1.25	91.3	0.735
R3	29.91	1.64	89.6	0.702

Whole-protein SASA range (all 4 replicas, PBC-corrected): 24–37 nm², mean 28.95 nm² — this is a different, larger quantity than the calyx-only SASA in §2.

Locked headline numbers (all PBC-corrected, verified across 8006 total frames):

- Total contact events: **613** (97 CENTER + 516 near-interface)
- Long events (≥ 10 ns): **6** — all non-activated contact
- Pearson r (SASA vs. orientation θ): **+0.006**, block-bootstrap 95% CI [**−0.09**, **+0.11**], effective $N \approx 17$ (SASA autocorrelation ≈ 232 ns) — rules out $|r| > 0.11$

Interpretation: The two-factor gate mechanism is disproven. A PBC artifact (`freesasa` without `mda.unwrap` inflates SASA to 45–62 nm²) originally suggested gate-opening events; after the fix, 0 gate-open events remain across all 8006 frames, and the SASA–orientation correlation is statistically indistinguishable from zero. Paper 1 is reframed around this as a scope claim: “we characterise the pre-commitment contact ensemble of native BLG at the air-water interface and define precisely where unbiased μ s MD stops” — not a discovered activation mechanism.

11 PCA (structural)

What & why: Principal component analysis of $C\alpha$ coordinates — summarizes the dominant modes of conformational variation over the CENTER trajectory.

Script: `scripts/analysis/blg_pca.py` (structural mode, wraps `gmx covar` → `gmx anaeig -proj`)

```
python -u scripts/analysis/blg_pca.py --mode structural --label CENTER
```

Output: `results/analysis/blg_pca_structural_CENTER.npz` — keys: `time_ns`, `pc1`, `pc2`, `variance_explained`, `eigenvalues`

Result: PC1 explains 35.1% of variance, PC2 13.4% (CENTER only). PC1 ranges -5.76 to $+5.09$ over the trajectory, centered at ≈ 0 .

12 Conformational clustering

What & why: Groups interface-contact frames by structural similarity (GROMOS algorithm, RMSD cutoff 0.2 nm on $C\alpha$) — reveals whether BLG samples one dominant conformation during contact or several distinct ones.

Script: `scripts/analysis/blg_cluster.py` (target: contact, wraps `gmx cluster`)

```
python -u scripts/analysis/blg_cluster.py --target contact --label CENTER
```

Output: `results/analysis/blg_cluster_contact_CENTER.npz` — keys: `frame_times_ns`, `cluster_ids`, `cluster_sizes`, `n_clusters`

Result: CENTER: **1 cluster**, all 220 contact frames (1.2–98.8 ns window) assigned to it — a single dominant conformation during this contact episode, no structural heterogeneity detected at the 0.2 nm cutoff.

13 Summary — all locked numbers in one place

These are the numbers CLAUDE.md marks as “never change without rerunning analysis” — the ones every downstream figure/claim in the paper must trace back to exactly these scripts and files.

Quantity	Value	Source
Total MD time	4.00 μ s (4×1000 ns)	§1
Contact events	613 (97 CENTER + 516 near-interface)	<code>blg_contact.py</code>
Long events (≥ 10 ns)	6 — all non-activated	<code>blg_verify_events.py</code>
Whole-protein SASA range	24–37 nm ² (mean 28.95)	<code>blg_gate_analysis.py</code> , §10
Calyx-only SASA (CENTER)	3.81 ± 0.46 nm ²	<code>blg_calyx_sasa.py</code> , §2
Pearson r (SASA vs. θ)	+0.006, 95% CI $[-0.09, +0.11]$	<code>blg_bootstrap.py</code> , §10
Effective N for r	≈ 17 (autocorr. ≈ 232 ns)	<code>blg_bootstrap.py</code>
R _g (CENTER)	1.504 ± 0.022 nm	<code>blg_rg.py</code> , §5
Patch RMSD (flat, CENTER/R2/R3)	0.20–0.28 nm	<code>blg_rmsd.py</code> , §3

Quantity	Value	Source
Patch RMSD (R1, elevated)	0.347 nm — real, contact-correlated	<code>blg_rmsd.py</code> , §3
Surface tension (TIP3P check)	≈ 52 mN/m, all 4 replicas	<code>blg_surface_tension.py</code> , §9

Known open items (not yet resolved, do not treat as final)

- `main.tex` line ≈ 144 still says TIP3P surface tension is “half” of 72 mN/m (≈ 35.8) — contradicts the verified 51.9 mN/m value above; needs a fix when paper work resumes.
- Hydrogen bonds (§7) have only been rerun with the corrected `gmx hbond` method for CEN-TER; R1/R2/R3 still pending.